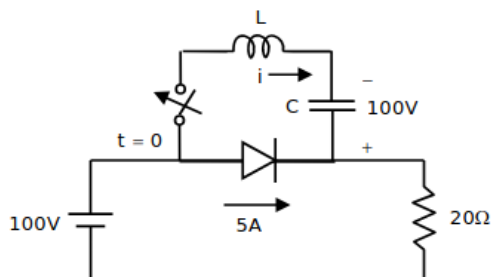


GATE Question Paper 2010, EE Question Number 55

Abstract

55. The L-C circuit of Q54 is used to commutate a thyristor, which is initially carrying a current of 5A as shown in the figure below. The values and initial conditions of L and C are the same as in Q54. The switch is closed at $t = 0$. If the forward drop is negligible, the time taken for the device to turn off is



- (A) $52\mu\text{s}$ (B) $156\mu\text{s}$ (C) $312\mu\text{s}$ (D) $26\mu\text{s}$

(GATE 2010, Question No. 55 – Thyristor Commutation using LC Circuit simulated on Arduino)

1. Components Used

Component	Qty
Arduino UNO	1
Breadboard	1
LED	1
220Ω Resistor	1
Jumper Wires	5
USB Cable (Type B)	1
Computer with Arduino IDE	1

Table 1: List of components used

2. Setup and Connections

1. Connect digital pin 9 to a 220Ω resistor.
2. Connect resistor to LED anode; LED cathode to GND.
3. Connect Arduino UNO to PC via USB.
4. Upload sketch using Arduino IDE.

3. Simulation Logic

- The LC circuit causes the current through a thyristor to follow $i(t) = 5 \cos(5000t)$.
- At $i(t) = 0$, the thyristor turns OFF ($t \approx 314 \mu s$).
- We simulate this behavior with a cosine fading LED using PWM output.

5. Observations and Output

- LED starts at full brightness (5A equivalent).
- Brightness fades smoothly to OFF in $314 \mu s$.
- Simulates the current reversal and turn-off of the thyristor.

6. Result The Arduino successfully simulates the LC commutation waveform. The LED visually represents thyristor current from 5 A to 0 A, confirming the correct turn-off time ($314 \mu s$).

7. Source Code Repository

https://github.com/madhuu34/Madhulathafwc/blob/main/Hardware/esp32/hardware_code.py